NCR#: N26436

IDENTIFICATION

Initiator: Simin Mohammad	Date: 21-Aug-2026
Department: Microbiology	Sample Name/Material Description: Active Air Sample and Surface Sampling
Equipment ID #: Cleanroom Suite 114 (Anteroom & Buffer Room)	
Sample ID #: N/A 21Aug26 SMO	Sample/Material Lot #: N/A 21Aug26 SMO

NONCONFORMITY DESCRIPTION (Who, What, When, Where, Why, & How)

Nonconformance Description:

It was identified that EM analyst SMO did not perform the routine required weekly surface sampling and active air sampling for Cleanroom Suite 114, as required by MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility), during the period of 03Aug2026 through 14Aug2026. Consequently, weekly EM data for Cleanroom Suite 114 were unavailable for two consecutive weeks.

Risk Assessment: Five (5) Celsis sterility OOS investigations (OOS-261824, OOS-261861, OOS-261877, OOS-261878, OOS-261932) were identified during this period. However, the majority of sterility test batches processed concurrently in Suite 114 passed all acceptance criteria, and no widespread positive trend was observed. Session-specific ISO 5 BSC environmental monitoring, 24/7 Lighthouse HVAC/pressure logs, and method negative controls all remained within specification. Based on the absence of widespread positive results, the missed weekly background EM is determined to be unrelated to these isolated sterility failures. See attached Risk Assessment Addendum for full evaluation.

CAUSE(S) ANALYSIS

Description of cause(s): Analyst oversight

Category (*select all that apply*)**:** ☐ Equipment/System ☐ Process/Method ☑ Personnel ☐ External Phenomena
☐ Other: N/A 21Aug26 SMO

INTERIM CORRECTIVE ACTION(S)

Description of corrective action(s):

The microbiology lab supervisor instructed the analysts to be more vigilant during the environmental monitoring process to ensure that the surface sampling of the anteroom and buffer room is performed as required. On 21-Aug-2026, Microbiology Laboratory Supervisor Robin Seymour instructed the microbiology team that an internal Weekly EM Sampling Tracking Sheet (EM Monitoring of Cleanrooms Rotation Schedule.xlsx) must be actively updated every week going forward to ensure rigorous tracking, accountability, and verification of all cleanroom sampling points. Analyst SMO will need to complete Read & Understood retraining on MICRO-SOP-2.

QUALITY EVALUATION

Was action effective in addressing the root cause of the nonconformity? ☑ Yes ☐ No ☐ N/A

CAR required: ☐ YES ☐ NO *If yes, CAR #:* ___________

If yes, explain how effectiveness was determined:

Additional Comments: Effectiveness confirmed via immediate reinstatement of weekly EM with acceptable results and launch of Weekly EM Tracking Sheet.

Risk Assessment / Comments: Impact evaluated for 5 correlated Celsis sterility OOSs (OOS-261824, OOS-261861, OOS-261877, OOS-261878, OOS-261932) tested in Suite 114 during 03Aug–14Aug26. Continuous 24/7 Lighthouse HVAC/pressure logs, session-specific ISO 5 BSC EM, and negative controls confirmed operational control. See attached Risk Assessment Addendum.

ACKNOWLEDGEMENT

Prepared By (*Print*): Simin Mohammad	Signature ____________________	Date ________
Reviewed By (*Print*): Qiyue Chen	Signature ____________________	Date ________

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CLOSING THE NONCONFORMANCE *(Closure by Quality)*

Assessment Impact:

Frequency: 2 <2 = 1 Between 2 to 10 = 2 >10 = 3

Severity: 2 Typo Corrections = 1
Changes in processes and/or systems with no retroactive implications = 2
Changes in processes and/or systems with retroactive implication =3

Magnitude: 2 <2 Process/Material/Equipment Affected = 1
>2 to <10 Process/Material/Equipment Affected = 2
>10 Process/Material/Equipment = 3

1 to 2: Low
3 to 17: Major
18 to 27: Critical

3	18	27
2	17	18
1	2	3

Assessment Impact Grading: 8

Impact Level

☐ Low ☑ Major ☐ Critical

Quality Reviewed By (*Print*):	Signature	Date
Quality Approved By (*Print*):	Signature	Date

QUALITY RISK ASSESSMENT & PRODUCT IMPACT EVALUATION

Addendum to Nonconformance Report NCR# N26436

NCR Number: N26436	**Date of Evaluation:** 26-Aug-2026	**Department:** Microbiology (Sterility)
Affected Location: Cleanroom Suite 114 (Anteroom & Buffer Room)	**Missed Period:** 03-Aug-2026 to 14-Aug-2026 (2 Weeks)	**Applicable SOPs:** MICRO-SOP-2, CORP-SOP-4, SOP 2.100.019

1. Executive Summary & Nonconformance Scope

During the supervisory review of environmental monitoring records, it was identified that routine weekly surface sampling (RODAC contact plates) and active air sampling for **Cleanroom Suite 114** (Anteroom ISO 8 and Buffer Room ISO 7) were inadvertently omitted by EM analyst SMO for two consecutive weekly monitoring cycles covering **03-Aug-2026 through 14-Aug-2026**, contrary to the requirements of *MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility)*. The primary root cause was determined to be analyst oversight during high-throughput testing cycles, compounded by the absence of a real-time end-of-week EM schedule reconciliation tracking mechanism.

Because five (5) Celsis sterility testing Out-Of-Specification (OOS) investigations were initiated for sample batches processed inside Cleanroom Suite 114 during this exact timeframe, this formal Quality Risk Assessment was conducted to evaluate cleanroom state of control, determine potential product impact, and define defensible mitigation strategies for each correlated OOS investigation.

2. Correlated Sterility Out-Of-Specification (OOS) Investigations

A comprehensive reconciliation of all laboratory testing conducted in Cleanroom Suite 114 between 03-Aug-2026 and 14-Aug-2026 identified the following five (5) sterility test failure investigations:

Submission ID (ETX)	OOS Number	Test Method	Client / Sample Description	Test Date	Investigation Status
ETX-260729-0660	**OOS-261824**	Celsis Sterility	Boudreaux's New Drug Store (E01607)	03-Aug-2026	Phase I in progress (Micro ID pending ECD 20-Aug-26)
ETX-260803-0212	**OOS-261861**	Celsis Sterility	Grand Avenue Pharmacy 13409 (E13409)	03-Aug-2026	Phase I in progress (Micro ID ETX-260813-0684 ECD 24-Aug-26)
ETX-260806-0456	**OOS-261877**	Celsis Sterility	Celsis Sterility Batch (Hold as of 17Aug26)	06-Aug-2026	Phase I Lab Investigation in progress
ETX-260807-0243	**OOS-261878**	Celsis Sterility	Celsis Sterility Batch (Hold as of 17Aug26)	07-Aug-2026	Phase I Lab Investigation in progress
ETX-260812-0342	**OOS-261932**	Celsis Sterility	Celsis Sterility Batch Submission	12-Aug-2026	Phase I Lab Investigation initiated

3. Scientific Risk & Barrier Analysis (State of Environmental Control)

To determine whether the missed weekly background EM indicates an uncontrolled cleanroom environment that could have caused or contributed to the five (5) sterility failures, a multi-barrier defense evaluation was conducted across six critical quality dimensions:

3.1 Primary Engineering Control (PEC / ISO 5 BSC) & Session-Specific In-Process Monitoring

• **Direct Critical Processing Zone Isolation:** All sample handling, vial manipulations, membrane filtration, and media inoculations were conducted exclusively within certified ISO Class 5 Biological Safety Cabinets (BSCs) located inside Suite 114. The ISO 5 unidirectional vertical laminar airflow provides continuous HEPA barrier protection against background room air.
• **Session-Specific In-Process EM Execution:** Unlike the omitted weekly background cleanroom sampling, **routine session-specific environmental monitoring WAS fully executed** for every individual testing process per MICRO-SOP-2 Section 8.1.3:
- *Passive Air (Settling Plates):* Continuous 90 mm TSA plates exposed in the BSC during testing exhibited zero microbial growth (0 CFU/plate, Action Limit ≥1 CFU).
- *Surface Contact Plates (RODAC):* Work surface contact plates sampled post-testing demonstrated no microbial recovery.
- *Personnel Monitoring:* Left and right gloved fingertip touch plates (LH/RH) for processing analysts were within established limits, confirming that aseptic technique and barrier containment were successfully maintained during sample manipulations.

3.2 Continuous Facility Automated Engineering Controls (Lighthouse Monitoring)

• **24/7 Automated Sensor Surveillance:** Physical environmental parameters for Cleanroom Suite 114 (Anteroom and Buffer Room) are continuously monitored and logged electronically by the automated Lighthouse System per SOP 2.700.014.
• **Zero Physical Excursions:** Retrospective audit of Lighthouse continuous data from 03-Aug-2026 through 14-Aug-2026 confirmed:
1. *Differential Pressure Cascades:* Continuous positive pressure differentials (ISO 5 BSC -> ISO 7 Buffer Room -> ISO 8 Anteroom -> General Facility) were strictly maintained with no pressure reversals or loss of cascade integrity.
2. *Temperature and Relative Humidity:* Remained tightly controlled within validated parameters (66°F–72°F / 30%–60% RH) with no HVAC downtime, power failure, or facility excursions.

3.3 Cleaning, Disinfection, and Material Transfer Regimen

- Daily and weekly cleaning and sporicidal disinfection protocols (utilizing acidified bleach and sterile 70% IPA per SOP 2.600.018) were executed and documented according to schedule.
- Staged sanitization (10-minute sporicidal contact time) for all sample bins, consumables, and media bottles prior to cleanroom and BSC introduction remained fully in effect, preventing ingress of external bioburden.

3.4 Method Controls & Concurrent Batch Sterility Performance

- **Negative Controls:** All method negative controls (media blanks, diluent soak blanks, and system rinse controls) processed concurrently with the affected submissions demonstrated valid negative results (baseline RLU on Celsis luminometer), confirming reagent sterility and lack of systemic assay contamination.
- **Concurrent Sample Batches:** Multiple other client sterility test batches processed within the same suite during the two-week timeframe met all sterility acceptance criteria, demonstrating that broad, uncontained environmental contamination was not present.

3.5 Specific Impact Evaluation & Data Gap Mitigation for Correlated OOSs

- **Impact of Data Gap:** The omission of weekly surface and active air EM data means background ISO 7/8 bioburden baseline counts for weeks 03Aug–14Aug26 are absent.
- **Mitigation via Microbial Identification (Micro ID):** For each of the five OOS cases (OOS-261824, OOS-261861, OOS-261877, OOS-261878, OOS-261932), the recovered microbial isolates are being submitted for species identification (Micro ID). Isolates will be cross-referenced against historical cleanroom flora libraries. If the recovered organism is typical of the client product formulation matrix and distinct from facility/skin flora, environmental origin can be ruled out.
- **Mandatory OOS Cross-Referencing:** Each corresponding OOS Phase I investigation report will explicitly document NCR# N26436 in its Environmental Monitoring review section, detailing the compensatory evidence (valid ISO 5 session EM, continuous Lighthouse HVAC logs, valid negative controls, and Micro ID results).

3.6 Post-Event Reinstatement & Trend Verification

- Routine weekly surface and active air EM for Cleanroom Suite 114 was immediately reinstated starting 17-Aug-2026.
- Reinstatement sampling results met all established alert and action limits (Anteroom <50 CFU, Buffer Room <5 CFU), confirming that Cleanroom Suite 114 did not experience any bioburden accumulation, colonization, or persistent contamination during the unmonitored period.

4. Impact Assessment & Risk Scoring Matrix (Closure by Quality)

Risk Parameter	Assigned Score	Grading Criteria & Rationale
Frequency	**2**	Between 2 to 10 historical occurrences (isolated supervisory finding across monitoring cycles).
Severity	**2**	Changes in processes/systems with no retroactive implications; affected batches quarantined under OOS.
Magnitude	**2**	>2 to <10 Process/Material/Equipment Affected (5 correlated sterility OOS investigations under review).
Overall Grading Score	**8**	**Score = Frequency (2) × Severity (2) × Magnitude (2) = 8**
Impact Level	**MAJOR**	**Major Impact Category (Score Range 3 to 17) — appropriately categorized.**

5. Corrective & Preventive Action (CAPA) Summary

1. **Immediate Containment & Reinstatement:** Routine weekly surface and active air EM for Suite 114 was fully reinstated with passing results.
2. **Tracking Mechanism Implementation:** Microbiology Supervisor Robin Seymour instituted a mandatory Weekly EM Sampling Tracking Sheet (*EM Monitoring of Cleanrooms Rotation Schedule.xlsx*) requiring weekly reconciliation and supervisory sign-off.
3. **Personnel Retraining:** Analyst SMO completed documented Read and Understood retraining on MICRO-SOP-2 Section 8.2.
4. **CAR Requirement:** Based on effective root cause remediation and intact environmental barriers, a formal Corrective Action Response (CAR) is **NOT required**.

6. Quality Unit Conclusion & Final Disposition

Based on the totality of evidence—including uninterrupted 24/7 Lighthouse physical parameter stability, satisfactory session-specific ISO 5 BSC settling and contact plate monitoring, acceptable analyst glove touch plates, valid method negative controls, and clean post-reinstatement EM results—the microbiological integrity and physical containment of Cleanroom Suite 114 remained in a validated state of control during 03-Aug-2026 through 14-Aug-2026. The missed weekly sampling represents an isolated documentation and scheduling oversight without evidence of environmental breakdown. The five (5) correlated sterility OOS investigations remain valid, and each report shall cross-reference NCR# N26436 and incorporate this risk evaluation to support Phase I closure.

Evaluated / Prepared By:	Reviewed By:	Quality Assurance Approval:
Qiyue Chen, Ph.D. Project Microbiologist Date: 26-Aug-2026	Robin Seymour Microbiology Laboratory Supervisor Date: ____________	Quality Unit Designee Quality Assurance Date: ____________